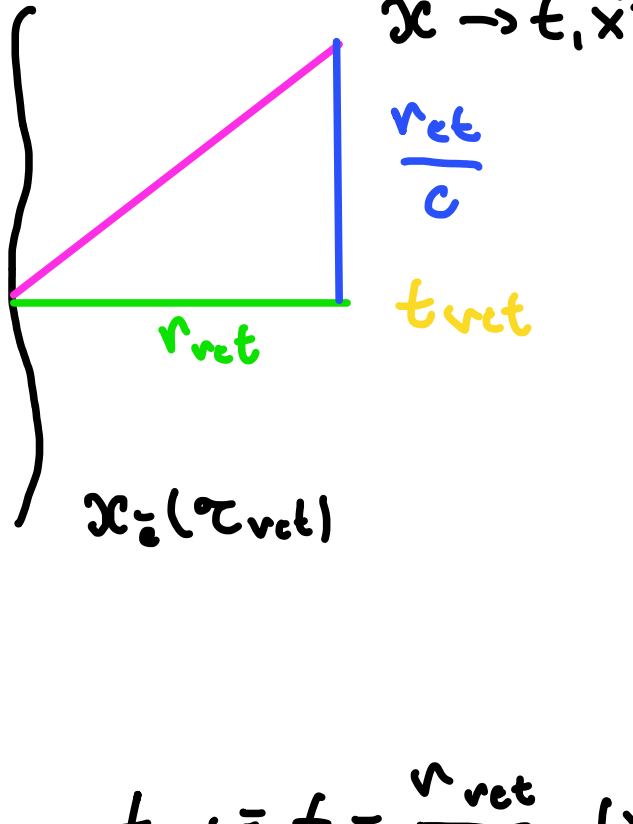


Relativita



$$\phi = \frac{q}{4\pi\epsilon_0} \frac{1}{r(1 - \frac{\vec{v} \cdot \vec{v}}{c^2})} \Big|_{t_{ret}}$$

$$\vec{A} = \frac{q}{4\pi\epsilon_0} \frac{\vec{v}}{r^2(1 - \frac{\vec{v} \cdot \vec{v}}{c^2})} \Big|_{t_{ret}}$$

$$\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$t_{ret} = t - \frac{r_{ret}}{c} (*)$$

$$c(t - t_{ret}) = r_{ret} = |\vec{x} - \vec{x}_0(t_{ret})|$$

$$\frac{1}{c} \frac{\partial}{\partial t} (*) = 1 - \frac{\partial t_{ret}}{\partial t} = \frac{1}{c} \frac{1}{r_{ret}} \vec{r}_{ret} \cdot \frac{\partial \vec{r}_{ret}}{\partial t}$$

$$= \frac{1}{c} \vec{e}_{ret} \cdot \vec{v}(t_{ret}) \frac{\partial t_{ret}}{\partial t}$$

$$\Rightarrow \frac{\partial t_{ret}}{\partial t} = \frac{1}{1 - \frac{\vec{v} \cdot \vec{v}}{c^2}} \Big|_{t_{ret}}$$

$$\vec{\nabla} (*) = -\vec{\nabla}(ct_{ret}) = \frac{1}{r_{ret}} (\vec{\nabla}(\vec{x} - \vec{x}_0(t_{ret})) \cdot \vec{r}_{ret})$$

$$= (\vec{I} - (\vec{\nabla} t_{ret}) \vec{v}) \cdot \vec{e}_{ret}$$

$$= \vec{e}_{ret} - (\vec{\nabla} t_{ret}) \vec{v} \cdot \vec{e} \Big|_{ret}$$

$$\Rightarrow \vec{\nabla} t_{ret} = - \frac{\vec{e}}{1 - \frac{\vec{v} \cdot \vec{v}}{c^2}} \Big|_{t_{ret}}$$

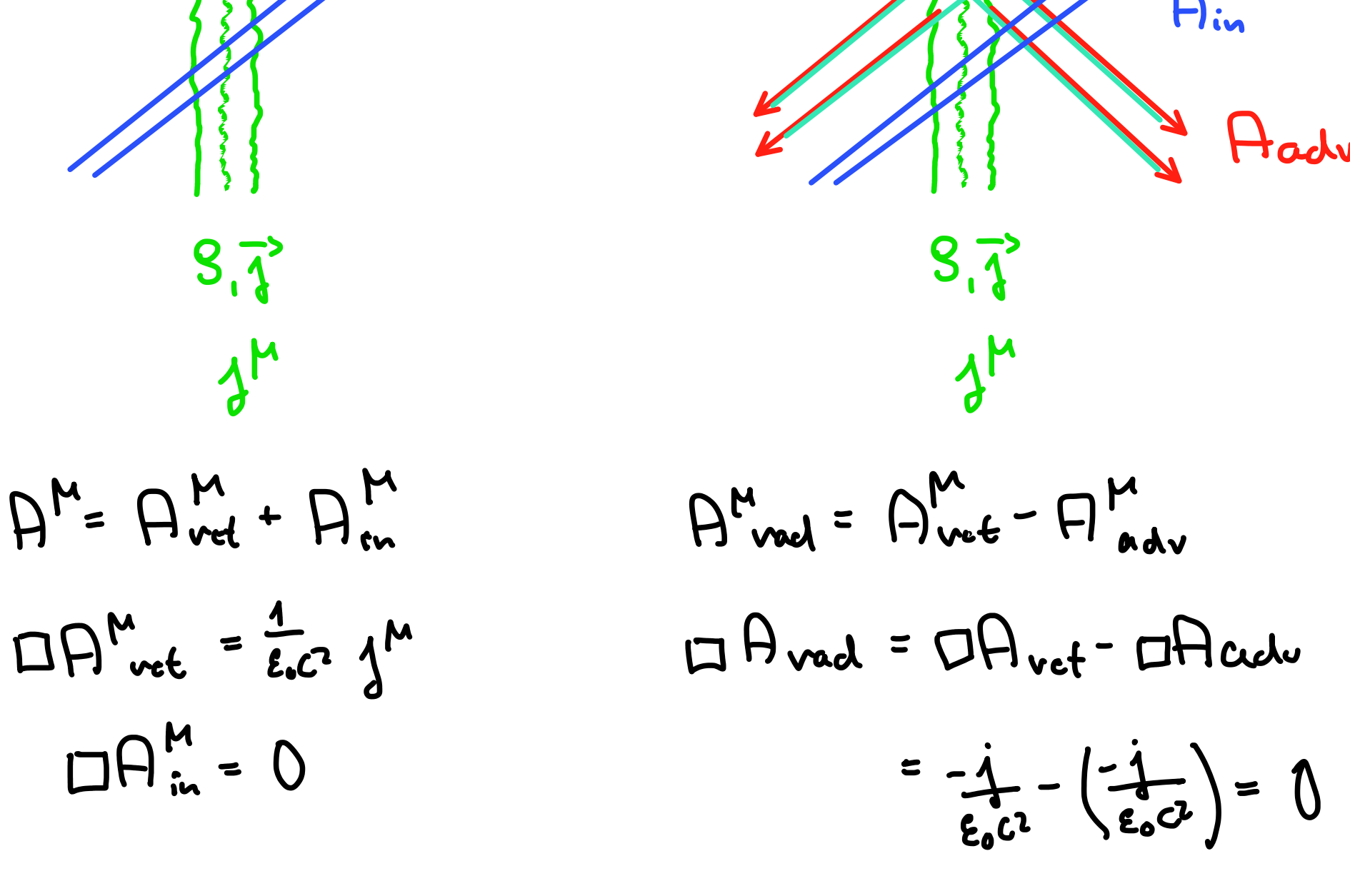
$$\Rightarrow \vec{\nabla} r_{ret} = \frac{\vec{e}}{1 - \frac{\vec{v} \cdot \vec{v}}{c^2}} \Big|_{t_{ret}}$$

$$\frac{\partial r_{ret}}{\partial t} = - \frac{\vec{e} \cdot \vec{v}}{1 - \frac{\vec{v} \cdot \vec{v}}{c^2}} \Big|_{t_{ret}}$$

$$\Rightarrow \vec{E} = \frac{q}{4\pi\epsilon_0} \frac{1}{(1 - \frac{\vec{v} \cdot \vec{v}}{c^2})^3} \left[\frac{\vec{e} - \frac{\vec{v}}{c}}{r^2} + \frac{\vec{e} \times ((\vec{e} - \frac{\vec{v}}{c}) \times \vec{v})}{c^2 r} \right]_{t_{ret}}$$

$$\vec{B} = \frac{1}{c} \vec{e}_{ret} \times \vec{E}$$

Distinze k retardovaním a odhadování řešení



$$A^M = A_{ret}^M + A_{in}^M$$

$$A_{ret}^M = A^M - A_{adv}^M$$

$$-\square A_{ret}^M = \frac{1}{\epsilon_0 c^2} j^M$$

$$\square A_{in}^M = 0$$

$$\square A_{adv}^M = \square A_{ret}^M - \square A_{adv}^M$$

$$= -\frac{1}{\epsilon_0 c^2} - \left(-\frac{1}{\epsilon_0 c^2}\right) = 0$$

$$A_{out} = A_{ret} - A_{adv} + A_{in}$$

$$= A_{ret} + A_{in}$$

→ uvažuj se nos klesání
⇒ předpokládá se na začátku
to bylo buď předpokládá
nebo počítat pole nejsou
klesání s budoucí
zdroji

$$A_{ret}^M(x) = \frac{1}{\epsilon_0 c^2} \int G_{ret}^M(x|x') j^M(x') d^4\Omega'$$

$$A_{adv}^M(x) = \frac{1}{\epsilon_0 c^2} \int (G_{ret}^M(x|x') - G_{adv}^M(x|x')) j^M(x') d^4\Omega'$$

$$\square G_c(x|x') = 0$$

$$G_c(x|x') = -G_c(x'|x)$$

$$\square A^M = 0 \quad \text{dává na } \Sigma \quad A^M(x) \quad \frac{\partial A^M}{\partial n}(x)$$

$$\int_{\Sigma} (\phi \square \eta - \eta \square \phi) d^4\Omega =$$

$$= - \int_{\Sigma_k} (\phi \frac{\partial \eta}{\partial n} - \eta \frac{\partial \phi}{\partial n}) d^3V + \int_{\Sigma_z} (\phi \frac{\partial \eta}{\partial n} - \eta \frac{\partial \phi}{\partial n}) d^3V$$

$$1) \quad x \in \langle \Sigma_k | \Sigma \rangle \quad \eta(x) \leftrightarrow A^M(x')$$

$$\phi(x') = G_{ret}(x|x')$$

$$A^M(x) = \int_{\langle \Sigma_k | \Sigma \rangle} \left[G_{ret}(x|x') \underbrace{\square' A^M(x')}_0 - A^M(x') \underbrace{\square' G_{ret}(x|x')}_{-\delta(x|x')} \right] d^4\Omega'$$

$$= \int_{x' \in \Sigma} \left[G_{ret}(x|x') \frac{\partial A^M}{\partial n'}(x') - A^M(x') \frac{\partial G_{ret}(x|x')}{\partial n'} \right] dV$$

$$- \int_{x' \in \Sigma_k} \left[\underbrace{G_{ret}}_0 \underbrace{\frac{\partial A^M}{\partial n'}}_{\text{konstanta struktura}} \right] dV$$

$$\Rightarrow \int_{x' \in \Sigma} \left[G_{ret}(x|x') \frac{\partial A^M}{\partial n'}(x') + \left(\frac{\partial G_{ret}}{\partial n'}(x|x') \right) A^M(x') \right] dV'$$

$$= \begin{cases} 0 & \times \text{ po } \Sigma \\ A^M(x) & \times \text{ před } \Sigma \end{cases}$$

$$\Rightarrow A^M(x) = \int_{x' \in \Sigma} \left[G_c(x|x') \frac{\partial A^M}{\partial n'}(x') + \left(\frac{\partial G_c}{\partial n'}(x|x') \right) A^M(x') \right] dV'$$

$$x, x' \in \Sigma$$

Brzdící síla

$$F = F_{ret} + F_{in} \quad \text{pole kolem náboje}$$

$$f = qF \cdot u = qF_{ret} \cdot u + qF_{in} \cdot u$$

→ symetrický příspěvek už nezapínej! ⇒ vyhodíme je,
zapínej! už s pouze nevyhodíme spotřebu s polárním,
ta působí na náboj

$$F_{ret} = F_{sym} + \frac{1}{2} F_{rad}$$

$$f_{self} = \frac{q}{2} F_{rad} \cdot u = \frac{q^2}{4\pi\epsilon_0} \left[\frac{1}{2} \frac{1}{3} \ddot{x}_0 + \frac{2}{3} (\ddot{x}_0^2 - \dot{x}_0^2 \ddot{x}_0) + O(s) \right]$$

$$M_0 \ddot{x}_0 = f_{self} + f_{ext}$$

$$\Rightarrow m_0 \ddot{x}_0 = f_{self} + f_{ext}$$

$$f_{self} = \frac{q^2}{4\pi\epsilon_0} \left[\frac{2}{3} (\ddot{x}_0^2 - \dot{x}_0^2 \ddot{x}_0) + O(s) \right]$$

$$m_0 = M_0 + \frac{1}{2} \frac{1}{3} \frac{q^2}{4\pi\epsilon_0}$$

$$\Rightarrow m_0 a = f_{ext} + \frac{2}{3} \frac{q^2}{4\pi\epsilon_0 m_0} m_0 (\ddot{a} - \dot{a}^2 u)$$

$$= f_{ext} + \tau_0 (\ddot{a} + u \ddot{a}) \cdot (m_0 a)^\perp$$

$$u^2 = -1$$

$$u \cdot a = 0$$

$$u \cdot \ddot{a} + \dot{a}^2 = 0$$

